

Souvik Barman

📍 Kolkata ✉ bsouvik337@gmail.com ☎ 7450876303

in <https://www.linkedin.com/in/souvik-barman-a19b36272/> 🌐 <https://github.com/bsouvik337>

Objective

Secure a challenging position in this digital world, where I can expertise in programming, algorithm design, and problem-solving to contribute to the creation of innovative and efficient software solutions. I am eager to apply my technical skills, coupled with a passion for problem solving and developing solutions.

Education

Central University of Kerala

Sept 2023 – May 2025

M.Sc. in Computer Science (specialization in intelligent systems)

- CGPA: 7.45/10.0

University of Calcutta

July 2019 – Aug 2022

B.Sc. in Physics (Hons.)

- CGPA: 7.503/10.0

Technical Skills

Languages: C++, C, SQL, Python

Developer Tools: Git, Firebase, Visual Studio, Jupyter Notebook, VS Code, Google Colab, MATLAB

Data Science: Data Cleaning, Preprocessing, Numpy, Pandas, Scikit-Learn

Model Framework: XGBoost, Neural Networks, NLP, Pytorch, Tensorflow, Gen AI, LightGBM

Web Technologies: HTML, CSS, react, nord.js

Machine Learning AI: Supervised learning, Unsupervised learning, Deep learning, Artificial intelligence

Working Details

Process Associate

Genpact, Kolkata

Python, Excel

7 July 2025 – Present

- Investigate high-risk user activity and suspicious transactions to identify and prevent fraudulent activity.
- Analyze trends, patterns, and behaviors in online transactions to detect fraud, abuse, and policy violations.
- Collaborate with cross-functional teams to enforce platform guidelines and improve risk mitigation strategies.
- Review and resolve user-reported issues while ensuring compliance with internal policies and regulatory standards.

AI/ML Intern

CAIR, DRDO

Google Colab, Python, MATLAB

24 Jan – 25 Apr 2025

- Experiments with BERT and Word2Vec based Document Clustering.
- The project evaluates different large language model (LLM) embeddings (BERT, Word2Vec) for the NLP task document clustering.
- The study assesses models using performance metric Cosine Similarity to determine their accuracy, efficiency, and scalability across different text types (news articles, research papers, blogs, etc.).
- The project follows a stepwise methodology, including literature review, dataset preparation, model implementation, performance benchmarking, and report documentation.
- The findings aim to provide insights into trade-offs between accuracy and computational efficiency, helping researchers and developers choose the best embedding techniques for NLP applications.

Machine Learning Intern

Hyderabad, Telangana

Google Colab, Python

1 May – 31 May 2024

- Fake News Detection using Machine Learning and Flask.
- The project successfully developed a machine learning-based fake news detection system with a highest accuracy of 95% using the Passive Aggressive Classifier.

Projects

Remote Health Monitoring Device (COVID Kit) ARM Cortex-M4, MQTT, sensor integration	25 Mar 2024
Credit Card Fraud Detection using machine learning techniques Google Colab, LR, SVM, DT	10 Dec 2023
Embedded ECG Monitoring Device TI microcontroller (MSP430), BLE module, RTOS	11 Nov 2023
Light Animations using Arduino and Matlab Arduino, MATLAB	12 Feb 2024
A Robust Heart Disease Prediction System Using Deep Neural Networks Machine Learning, Deep Learning, Python, MATLAB	25 Jul 2024 - 10 Jan 2025
Image Classification using Pretrained CNN model - VGG 16 Deep Learning, Python, MATLAB	27 Mar 2024
Implementation of denoising auto encoder on MNIST handwritten digit dataset Deep Learning, Python, Image Data	27 Mar 2024
Music Recommendation System Using Machine Learning Deep Learning, Python, Machine Learning	28 Feb 2024
Implement the Newton-Raphson method in MATLAB MATLAB, Python, Machine Learning, LR	6 Nov 2023
Matrix Transposition using OpenMP for Parallel Processing OpenMP, Python, C	15 Sept 2024
Atomic Operation and Barrier Synchronization in OpenMP OpenMP, Python, C	1 Sept 2024
Matrix Multiplication using OpenMP for Parallel Processing OpenMP, Python, C	9 Sept 2024

Certifications

- Microsoft Foundation AI and Machine Learning
- Tata Gen AI Powered Data Analytics Job Simulation.
- Google Analytics Certification.
- Udemy Master Class on Data Science using Python A-Z for ML.
- Use GA4 with other Tools and Data Sources.
- Career Essentials in Data Analysis by Microsoft and LinkedIn.
- Udemy Generative AI : Create an impressive AI Art 2025.

Achievements

- Qualified TCS NQT in the year 2022 with 70.35%.